

Chapter 8, Problem 5RQ

Problem

In a series RLC circuit, setting $R = 0$ will produce:

- (a) an overdamped response
- (b) a critically damped response
- (c) an underdamped response
- (d) an undamped response
- (e) none of the above

Step-by-step solution

Step 1 of 1

In a series RLC circuit, the damping in the response is due to the circuit element resistor, R . By adjusting the value of resistor, R , the response can be changed to overdamped, underdamped, critically damped or undamped response. The damping effect depends on the damping factor, α . Setting $R = 0$ will make $\alpha = 0$, and produce an **undamped response**.

Therefore, the correct option is **(d)**.